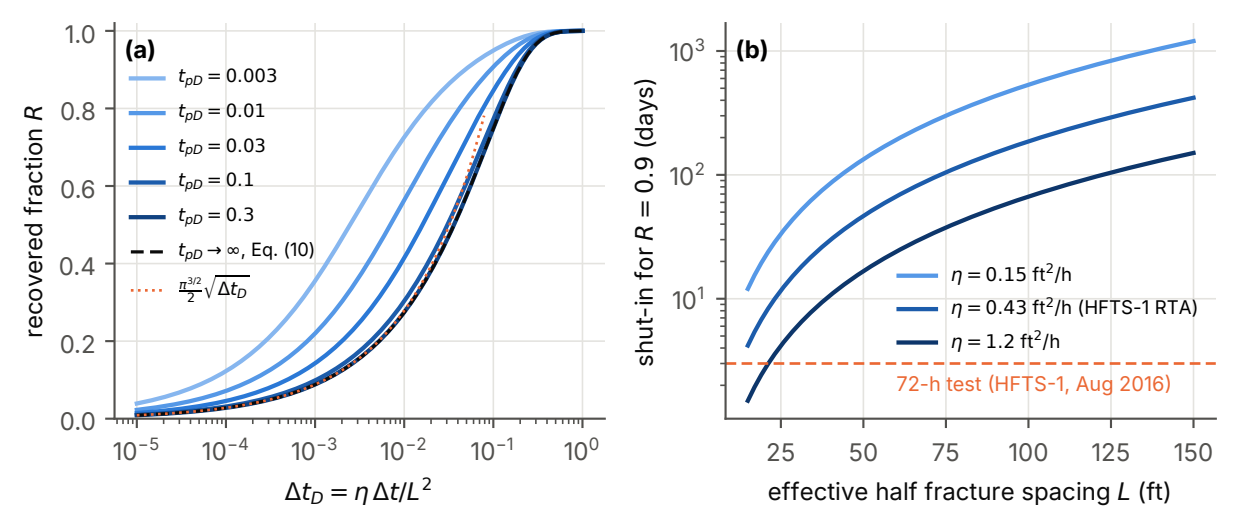
(a)
recovered fraction R
1.0
0.8
0.6
0.4
0.2
0.0
$t_{pD} = 0.003$
$t_{pD} = 0.01$
$t_{pD} = 0.03$
$t_{pD} = 0.1$
$t_{pD} = 0.3$
$t_{pD} \to \infty$, Eq. (10)
$\frac{\pi^{3/2}}{2}\sqrt{\Delta t_D}$
$10^{-5}$
$10^{-4}$
$10^{-3}$
$10^{-2}$
$10^{-1}$
$10^{0}$
$\Delta t_D = \eta\, \Delta t / L^2$
(b)
shut-in for $R = 0.9$ (days)
$10^{3}$
$10^{2}$
$10^{1}$
$\eta = 0.15$ ft$^2$/h
$\eta = 0.43$ ft$^2$/h (HFTS-1 RTA)
$\eta = 1.2$ ft$^2$/h
72-h test (HFTS-1, Aug 2016)
25
50
75
100
125
150
effective half fracture spacing $L$ (ft)